

Liu’s Second Hypothesis Holds, and the Union-Closed Constant $c' = 1 - m^* = 0.38270908791873502993 \dots$ Is Unconditional: a Machine-Checked Proof

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Abstract

Liu (arXiv:2306.08824) showed that the union-closed sets conjecture holds with constant $c' = 1 - p^*x^* \approx 0.38271$ under two hypotheses: a positive-semidefiniteness statement (Section V-A) and the assertion that a nine-parameter optimisation attains its global minimum at an explicit two-point law (Section V-B, “Hypothesis 2”). We prove that the two-protocol entropy inequality underlying Hypothesis 2 holds for *every* pair of Borel probability measures on $[0, 1]$ and, more generally, for every conditionally i.i.d. coupling, with the uniform constant M/m^* and no mean constraint. The inequality Liu’s Proposition 3 needs therefore holds outright, both hypotheses are bypassed, and c' is an unconditional lower bound for the union-closed constant. Exact identities in the algebra of pairings reduce the proof to a sufficient pair of inequalities between explicit continuous kernels on the unit square, $A \geq 0$ and $B \geq 0$; $A \geq 0$ (together with the tight inequality $R = A + B \geq 0$, which has a nondegenerate interior zero) is certified by stratified analytic estimates and a finite Arb branch-and-bound cover, while $B \geq 0$ is equivalent to a parameter-free inequality $\Phi \geq 0$ that follows from an exact identity and a one-variable condition $\kappa \geq 2$ with uniform margin. Every non-elementary step is a machine-checked artifact with pinned hashes; a replay driver re-runs the seven artifacts and asserts byte-identity. The bound is sharp along a one-parameter family, and Liu’s printed decimal 0.382709087918741 is incorrect in its last two digits. The result concerns the constant only and says nothing about the conjectured value $1/2$.

1 Introduction and main results

Let $\mathcal{F}$ be a union-closed family of subsets of a finite set E , with $\mathcal{F} \neq \emptyset$ and $\mathcal{F} \neq \{\emptyset\}$. Frankl’s conjecture asserts that some element of E belongs to at least half of the members of $\mathcal{F}$. Gilmer [4] proved the first constant bound by an entropy argument; the sharp form of his inequality gives $(3 - \sqrt{5})/2$ [3, 1, 7], Sawin’s max-entropy coupling improves it, and the best value obtainable along that route, $c^* \approx 0.3823455$, was computed by Yu [8] and Cambie [2]. Liu [6] introduced conditionally i.i.d. couplings and showed that, with the specific coupling of his Example 5 (with $f(x) = x(1 - x)$) mixed with Gilmer’s independent coupling, the constant improves to

$$c' = 1 - m^*, \quad m^* = p^*x^*, \tag{1}$$

*Repository manuscript package: `math/uc/H2.PAPER/`. Every theorem carries a status label (PROVED / MACHINE-VERIFIED / COMPUTATIONAL-EVIDENCE / OPEN), and every machine-assisted theorem cites the artifact and SHA-256 that certifies it; Section 10 explains how to replay the whole chain.

where x^* and p^* are the root-defined numbers of his equations (87)–(88) (Section 2), *provided* two hypotheses hold (his Theorem 13): the positive semidefiniteness of Section V-A (“Hypothesis 1”), which drives a Krein–Milman reduction of the admissible couplings to three shared-mass atoms, and the global-minimiser structure of Section V-B (“Hypothesis 2”), which asserts that the resulting nine-parameter optimisation (his (81)–(84)) has optimal value at least 1.

This paper proves the inequality behind Hypothesis 2 in a form strong enough to make Hypothesis 1 unnecessary. Throughout, $\log$ is natural,

$$h(u) = -u \log u - (1-u) \log(1-u), \quad h(0) = h(1) = 0, \quad \pi(s, t) = st(1 + (1-s)(1-t)), \quad K = h \circ \pi. \quad (2)$$

For a Borel probability measure μ on $[0, 1]$ write $M = \langle \mu, x \rangle$ for its mean and $\langle \mu, h \rangle = \int h d\mu$.

Definition 1.1 (conditionally i.i.d. coupling). Let (Ω, P) be a probability space and $u \mapsto \nu_u$ a Markov kernel from Ω to $[0, 1]$. The measure $\Gamma = \int \nu_u \otimes \nu_u dP(u)$ on $[0, 1]^2$ is a *conditionally i.i.d. coupling* of its barycentre $\mu = \int \nu_u dP(u)$ with itself. Its *numerator* at the parameter $\beta \in (0, 1)$ is

$$\text{num}(\Gamma) = (1 - \beta) \langle \mu \otimes \mu, h(xy) \rangle + \beta \int \langle \nu_u \otimes \nu_u, K \rangle dP(u). \quad (3)$$

Theorem 1.2 (mixture theorem; PROVED, Theorem 7.1). *Let $\beta = \beta^*$ and $m = m^*$ be the constants of Section 2. For every Borel probability measure μ on $[0, 1]$ and every conditionally i.i.d. coupling Γ of μ ,*

$$\text{num}(\Gamma) \geq \frac{M}{m^*} \langle \mu, h \rangle.$$

The inequality is an equality along the whole family $\mu_w = (1-w)\delta_0 + w\delta_{x^}$, $0 \leq w \leq 1$, $\Gamma = \mu_w \otimes \mu_w$.*

For the two-component couplings $\Gamma = (1-q)\mu \otimes \mu + q\nu \otimes \nu$ this is Liu’s two-protocol objective with the mean constraint removed:

Theorem 1.3 (Hypothesis 2 for general measures; PROVED, Theorem 6.1). *For all Borel probability measures μ, ν on $[0, 1]$ and all $q \in [0, 1]$, with $P = (1-q)\mu + q\nu$ and $M = \langle P, x \rangle$,*

$$\text{gap}(\mu, \nu, q) := (1-\beta^*) \langle P \otimes P, h(xy) \rangle + \beta^* [(1-q) \langle \mu \otimes \mu, K \rangle + q \langle \nu \otimes \nu, K \rangle] - \langle P, h \rangle \geq \frac{M - m^*}{m^*} \langle P, h \rangle,$$

so $\text{gap} \geq 0$ whenever $M \geq m^$. Liu’s nine-parameter optimisation (81)–(84) at $c = c'$ is the case of finitely supported μ, ν with three shared masses; its optimal value is therefore ≥ 1 , i.e. Hypothesis 2 holds, with equality at his minimiser $q = 0$, $\mu = (1-p^*)\delta_0 + p^*\delta_{x^*}$.*

Theorem 1.4 (the constant is unconditional; PROVED, Theorem 8.3). *Every union-closed family $\mathcal{F} \not\subseteq \{\emptyset, \{\emptyset\}\}$ of subsets of a finite set has an element contained in at least $c' |\mathcal{F}|$ of its members, where*

$$c' = 1 - m^* \in [0.382709087918735029930312902098972611626381433 \pm 4.95 \times 10^{-46}].$$

The proof of Theorem 1.4 re-derives Liu’s Proposition 3 from his definitions (Section 8); the only inputs are Gilmer’s independent protocol and Liu’s Example 5, and Theorem 1.2 applied coordinate by coordinate to the couplings those protocols actually induce. Liu’s Theorem 9 (two-mixture sufficiency), Lemma 11 and Theorem 12 (the Krein–Milman reduction, which is where Hypothesis 1 enters), and Section V-A are not used. Theorem 1.2 also supplies the strict constant $C = (1 - c)/m^* > 1$ that Proposition 3 requires for every $c < c'$, with no limiting argument.

How the proof is organised. Section 2 fixes notation and proves the exact structure of the constants: x^* is the unique root in $(0, 1)$ of $z^4 - 2z^3 + 3z^2 - 1$, which is precisely the relation $\pi(x, x) = 1 - x^2$, so that $K(x^*, x^*) = h(x^{*2})$ and the kernel $R = D_{m^*}/m^*$ vanishes to second order at (x^*, x^*) . Section 3 proves the reduction: three identities in the algebra of pairings express gap through the kernels

$$P_2, Q_2, R = P_2 + Q_2, \varphi = \sqrt{Q_2(s, s)}, A = P_2 + \varphi \otimes \varphi, B = Q_2 - \varphi \otimes \varphi,$$

and show that $A \geq 0$ and $B \geq 0$ pointwise on $[0, 1]^2$ imply Theorem 1.3. Section 4 proves $R \geq 0$ and $A \geq 0$ by stratified analytic estimates plus a finite certified cover. Section 5 proves the parameter-free inequality $\Phi = K_{st}^2 - K_{ss}K_{tt} \geq 0$, which is equivalent to $B \geq 0$, from an exact identity and a one-variable condition $\kappa \geq 2$. Section 6 composes these into Theorem 1.3, and Section 7 proves Theorem 1.2. Section 8 proves Theorem 1.4, Section 9 discusses the value of c' , and Section 10 lists the artifacts, their hashes, the replay procedure, and the scope of the result.

Labels. PROVED means a complete argument is given here or in the cited artifact's certificate; MACHINE-VERIFIED means an identity or inequality checked by exact symbolic algebra or by Arb ball arithmetic with the stated precision, on the stated set; COMPUTATIONAL-EVIDENCE means a floating-point or sampled observation that carries no weight in any proof; OPEN means not established. Nothing labelled COMPUTATIONAL-EVIDENCE is used in the proofs.

Pairings. For finite signed Borel measures α, β on $[0, 1]$ and bounded Borel k on $[0, 1]^2$ we write $\langle \alpha \otimes \beta, k \rangle = \iint k(s, t) d\alpha(s) d\beta(t)$ and $\langle \alpha, f \rangle = \int f d\alpha$. Three facts are used throughout: bilinearity; symmetry, $\langle \alpha \otimes \beta, k \rangle = \langle \beta \otimes \alpha, k \rangle$ whenever $k(s, t) = k(t, s)$; and the product rule $\langle \alpha \otimes \beta, f(s)g(t) \rangle = \langle \alpha, f \rangle \langle \beta, g \rangle$. With subscripts 0, 1 for μ, ν we abbreviate $E_{ab} = \langle \cdot, h(xy) \rangle$, $K_{ab} = \langle \cdot, K \rangle$, $H_a = \langle \cdot, h \rangle$, $M_a = \langle \cdot, x \rangle$, $F_a = \langle \cdot, \varphi \rangle$.

2 Notation, the constants, and the structure at (x^*, x^*)

Lemma 2.1 (regularity; PROVED). *h is continuous on $[0, 1]$, $0 \leq h \leq \log 2$, $h(u) = h(1 - u)$, and π maps $[0, 1]^2$ into $[0, 1]$. Consequently every kernel appearing in this paper ($h(xy)$, K , D_M , P_2 , Q_2 , R , φ , A , B , Φ) is continuous on the compact square, and every pairing below is finite.*

Proof. $h''(u) = -1/(u(1 - u)) < 0$, $h(1/2) = \log 2$, $u \log u \rightarrow 0$, and the symmetry is visible from the definition. With $a = 1 - s$, $b = 1 - t$, $1 - \pi(s, t) = a(1 - b) + b(1 - a) + ab(a + b - ab)$, a sum of nonnegative terms, and $\pi \geq 0$; so $\pi \in [0, 1]$. Compositions and products of continuous functions are continuous, and $\varphi = \sqrt{\beta K(s, s)}$ is continuous because $K(s, s) \geq 0$. $\square$

Definition 2.2 (the constants; Liu (87)–(90)). Let $f(z) = z^4 - 2z^3 + 3z^2 - 1$ and let x^* be its unique root in $(0, 1)$. Put

$$p^* = \frac{h(x^*)}{h(x^{*2})}, \quad m^* = p^* x^*, \quad c' = 1 - m^*, \quad \beta^* = \frac{(h'(x^*) + h(x^*)/x^*)/p^* - a'}{b' - a'},$$

where $a' = 2x^*h'(x^{*2})$, $b' = h'(\pi(x^*, x^*))\pi'_d(x^*)$, $\pi_d(x) = \pi(x, x) = x^2(2 - 2x + x^2)$ and $\pi'_d(x) = 2x(2 - 3x + 2x^2)$.

Lemma 2.3 (PROVED; enclosures MACHINE-VERIFIED).

(a) *f has exactly one root in $(0, 1)$: $f(0) = -1 < 0 < f(1) = 1$ and $f'(z) = 2z(2(z - \frac{3}{4})^2 + \frac{15}{8}) > 0$ on $(0, 1]$.*

- (b) $f(x) = 0$ is the relation $x^2 + \pi(x, x) = 1$, i.e. Liu's (87). Hence $\pi(x^*, x^*) = 1 - x^{*2}$ and $K(x^*, x^*) = h(x^{*2})$.
- (c) Liu's (88), $p^2 h(x^2) - ph(x) = 0$ with $p \neq 0$, is $p^* = h(x^*)/h(x^{*2})$; and β^* is the unique β for which the stationarity conditions (89)–(90) hold at (p^*, x^*) , equivalently (Lemma 2.4) for which $\frac{d}{dx} R_\beta(x, x)|_{x^*} = 0$.
- (d) Certified enclosures (Arb, 400 bits, propagated from a rational bracket of x^* of width 10^{-70} on which $f(\text{lo}) < 0 < f(\text{hi})$ is certified):

$$\begin{aligned} x^* &\in [0.690787593924988014150488204636270562580628177 \pm 9.69 \times 10^{-47}], \\ p^* &\in [0.893604513905465446476770019038769597178677710 \pm 3.49 \times 10^{-46}], \\ m^* &\in [0.617290912081264970069687097901027388373618567 \pm 4.95 \times 10^{-46}], \\ \beta^* &\in [0.100052559862893106664628299476947587796206185 \pm 2.88 \times 10^{-46}], \\ c' &\in [0.382709087918735029930312902098972611626381433 \pm 4.95 \times 10^{-46}]. \end{aligned}$$

Proof. (a) is the displayed computation. (b): $\pi(x, x) = x^2(1 + (1 - x)^2) = 2x^2 - 2x^3 + x^4$, so $x^2 + \pi(x, x) - 1 = f(x)$; then $h(1 - x^{*2}) = h(x^{*2})$ by the symmetry of h . (c): the objective at the two-point law $p\delta_x + (1 - p)\delta_0$ with $q = 0$ is $G(p, x) = (1 - \beta)p^2 h(x^2) + \beta p^2 h(\pi_d(x)) - ph(x)$, and (89)–(90) ask that $dG = 0$ along $d(px) = 0$, i.e. $dp = -(p/x)dx$. Writing out $-(p/x)\partial_p G + \partial_x G = 0$, using $h(\pi_d(x^*)) = h(x^{*2})$ and $p^* h(x^{*2}) = h(x^*)$, one obtains $(1 - \beta)a' + \beta b' = (h'(x^*) + h(x^*)/x^*)/p^*$, whose unique solution is the displayed β^* . (d) is the content of the `constants` blocks of the artifacts of Sections 4, 6 and 7, which agree with one another and with the independent solver `uc/liu9_binding.py` (difference enclosures containing 0). $\square$

From now on $\beta = \beta^*$ and $m = m^*$ unless a formula displays β, m as free parameters. Define, for $(s, t) \in [0, 1]^2$ and $M > 0$,

$$D_M(s, t) = M[(1 - \beta)h(st) + \beta K(s, t)] - \frac{1}{2}(th(s) + sh(t)), \quad (4)$$

$$P_2(s, t) = (1 - \beta)h(st) - \frac{th(s) + sh(t)}{2m}, \quad Q_2(s, t) = \beta K(s, t) \geq 0, \quad R = P_2 + Q_2 = \frac{D_m}{m}, \quad (5)$$

$$\varphi(s) = \sqrt{Q_2(s, s)}, \quad A = P_2 + \varphi \otimes \varphi, \quad B = Q_2 - \varphi \otimes \varphi, \quad R = A + B, \quad (6)$$

$$\Phi(s, t) = K(s, t)^2 - K(s, s)K(t, t). \quad (7)$$

P_2 is sign-indefinite (numerically its minimum on the square is ≈ -0.0693 , on the diagonal near $s = t \approx 0.6807$; COMPUTATIONAL-EVIDENCE, not used); $Q_2 \geq 0$; Φ contains no parameter.

Lemma 2.4 (the tight point; PROVED, Hessian MACHINE-VERIFIED). $R(x^*, x^*) = 0$ and $\nabla R(x^*, x^*) = 0$; the Hessian of R at (x^*, x^*) has certified entries $R_{ss} = R_{tt} \in [0.75319572205793855886 \pm 10^{-20}]$, $R_{st} \in [-0.49637762911066539838 \pm 10^{-20}]$ (the artifact's radii are $\sim 10^{-51}$; every enclosure quoted in this paper is the artifact's ball rounded to the printed digits with the radius enlarged to cover the rounding), hence $\lambda_{\min} \geq 0.25681809294727316048$ and the zero is a nondegenerate interior minimum. Likewise $A(x^*, x^*) = 0$ and $\nabla A(x^*, x^*) = 0$, since $A(s, s) = R(s, s)$ on the diagonal and $\partial_s A(s, s) = \partial_s R(s, s)$ for $0 < s < 1$; the Hessian of A there is different, $A_{ss} = A_{tt} \in [0.59340358088849405417 \pm 10^{-20}]$, $A_{st} \in [-0.33658548794122089368 \pm 10^{-20}]$, but has the same certified $\lambda_{\min} \geq 0.25681809294727316048$ (the off-diagonal shift $\varphi'(x^*)^2$ cancels in $A_{ss} + A_{st}$). Moreover $R(0, t) = R(s, 0) = 0$ and $R(1, t) = (1 - \frac{1}{2m})h(t)$ with $1 - \frac{1}{2m} \in [0.19000913473001832145 \pm 10^{-20}]$.

Proof. On the diagonal, $r(x) := R(x, x) = (1-\beta)h(x^2) + \beta h(\pi_d(x)) - x h(x)/m$. At x^* , Lemma 2.3(b) gives $(1-\beta)h(x^{*2}) + \beta h(\pi_d(x^*)) = h(x^{*2})$ for every β , and $x^* h(x^*)/m^* = h(x^{*2})$ by the definition of m^* ; so $r(x^*) = 0$. Next $r'(x) = (1-\beta)2xh'(x^2) + \beta h'(\pi_d)\pi_d' - (h(x) + xh'(x))/m$, and at x^* , with $m^* = p^*x^*$, the condition $r'(x^*) = 0$ is exactly $(1-\beta)a' + \beta b' = (h'(x^*) + h(x^*)/x^*)/p^*$, which defines β^* . Since R is symmetric, $R_s = R_t$ on the diagonal and $r' = R_s + R_t = 2R_s$, so $\nabla R(x^*, x^*) = 0$. For A : $A(s, s) = P_2(s, s) + \varphi(s)^2 = R(s, s)$, and $\partial_s A(s, s) = \partial_s P_2(s, s) + \varphi'(s)\varphi(s) = \partial_s P_2(s, s) + \frac{1}{2}\frac{d}{ds}Q_2(s, s) = \partial_s P_2(s, s) + \partial_s Q_2(s, s)$ by symmetry of Q_2 , valid for $0 < s < 1$ where $\varphi > 0$ is smooth. The faces: $h(0) = 0$, $\pi(0, t) = 0$, $\pi(1, t) = t$. The Hessian entries and the face constant are Arb enclosures evaluated at the ball of x^* (artifact `liu9-h2-two-var.json`). $\square$

Remark 2.5 (why the constants are exact objects). Because $R \geq 0$ (Theorem 4.1) is *tight* at (x^*, x^*) with a positive-definite Hessian, the inequality is a statement about the exact (β^*, m^*) : lowering m makes $R(x^*, x^*)$ negative to first order in the perturbation, and moving β at fixed m leaves $R(x^*, x^*) = 0$ (Lemma 2.4) but destroys the stationarity, so R becomes negative near (x^*, x^*) (to second order). A first version of the artifact of Section 6 declared 100-digit rational truncations of the constants to be “the constants”; at those rationals Liu’s minimiser has a *certified negative* gap, $[-5.9548013684 \times 10^{-102} \pm 10^{-112}]$ (witness W1b of `liu9-h2-general-lift.json`). All sign claims in this paper are therefore evaluated at Arb balls that contain the exact constants, and the enclosures are how those constants are represented, never a licence to perturb them.

3 The reduction: identities in the algebra of pairings

Let μ, ν be Borel probability measures on $[0, 1]$, $q \in [0, 1]$, $P = (1-q)\mu + q\nu$, $M = \langle P, x \rangle = (1-q)M_0 + qM_1$, and assume $M > 0$ (if $M = 0$ then $P = \delta_0$ and gap = 0). Let

$$I_{ab} = \langle \cdot \otimes \cdot, D_M \rangle \quad (a, b \in \{0, 1\}), \quad c = \beta \langle (\mu - \nu) \otimes (\mu - \nu), K \rangle = \beta(K_{00} + K_{11} - 2K_{01}),$$

and $T = 2I_{01}/M + c$.

Theorem 3.1 (reduction; PROVED, MACHINE-VERIFIED).

$$(C1) \quad \text{gap}(\mu, \nu, q) = \frac{(1-q)^2 I_{00} + q^2 I_{11}}{M} + q(1-q)T \text{ for every } q \in [0, 1], \text{ with no constraint on } M.$$

$$(C4) \quad \frac{D_M}{M} - \frac{D_m}{m} = \frac{(th(s) + sh(t))(M - m)}{2Mm} \geq 0 \text{ pointwise whenever } M \geq m > 0.$$

$$(C5) \quad \text{At } M = m: T_m := c + 2\langle \mu \otimes \nu, R \rangle.$$

$$(C6) \quad T_m = \langle \mu \otimes \mu, B \rangle + \langle \nu \otimes \nu, B \rangle + 2\langle \mu \otimes \nu, A \rangle + (F_0 - F_1)^2.$$

Certificate. `liu9-h2-reduction.json`, sha256 `c63ae94e4fbd2397e19ea9e2a4e2c155f1e02265852d0aacad5c97d70f078e3a` (Arb, 480 bits: (C1) against Liu’s audited Matlab transcription `uc/liu9_objective.py` on the nine-variable class, (C4)–(C6), and the two nine-variable statements (C2) $T = a^T M T a$ and (C3) $\Psi_M(s, t, s, t) = 2D_M(s, t)/M$ used only in Remark 3.3; four mutations caught); independently re-derived in a different free-log basis with exact zero residuals and 4/4 mutations in `liu9-psi-reduction.json`, sha256 `9fb5766f1c3110cce9b231d2f2f1b7c1eec67bc3d6ac7602be50fc72b00c07a3`; (C1), (C6) as sympy identities with β, m free symbols in `liu9-h2-general-lift.json` (claims L1, L3).

Proof. (C1). Expand both sides in the pairings. $\langle P \otimes P, h(xy) \rangle = (1-q)^2 E_{00} + 2q(1-q)E_{01} + q^2 E_{11}$ and $\langle P, h \rangle = (1-q)H_0 + qH_1$. By the product rule, $\langle \mu \otimes \nu, th(s) + sh(t) \rangle = M_1 H_0 + M_0 H_1$, so $I_{01} = M[(1-\beta)E_{01} + \beta K_{01}] - \frac{1}{2}(M_1 H_0 + M_0 H_1)$, $I_{00} = M[(1-\beta)E_{00} + \beta K_{00}] - M_0 H_0$, and similarly I_{11} . On the right-hand side of (C1) the E -terms assemble to $(1-\beta)\langle P \otimes P, h(xy) \rangle$; the K -terms

are $\beta[(1-q)^2 K_{00} + q^2 K_{11} + 2q(1-q)K_{01}] + q(1-q)\beta[K_{00} + K_{11} - 2K_{01}] = \beta[(1-q)K_{00} + qK_{11}]$; and the H -terms are $-\frac{1}{M}[(1-q)H_0((1-q)M_0 + qM_1) + qH_1(qM_1 + (1-q)M_0)] = -(1-q)H_0 - qH_1$. This is gap.

(C4) is immediate from (4): the bracketed term cancels and $\frac{1}{2}(th(s) + sh(t))(\frac{1}{m} - \frac{1}{M}) \geq 0$.

(C5). $2I_{01}/m = 2\langle \mu \otimes \nu, D_m/m \rangle = 2\langle \mu \otimes \nu, R \rangle$.

(C6). $c + 2\langle \mu \otimes \nu, R \rangle = \beta(K_{00} + K_{11}) - 2\beta K_{01} + 2\langle \mu \otimes \nu, P_2 \rangle + 2\beta K_{01} = \beta K_{00} + \beta K_{11} + 2\langle \mu \otimes \nu, P_2 \rangle$.

On the other hand, by the product rule $\langle \mu \otimes \mu, \varphi \otimes \varphi \rangle = F_0^2$ etc., so $\langle \mu \otimes \mu, B \rangle + \langle \nu \otimes \nu, B \rangle + 2\langle \mu \otimes \nu, A \rangle + (F_0 - F_1)^2 = \beta K_{00} - F_0^2 + \beta K_{11} - F_1^2 + 2\langle \mu \otimes \nu, P_2 \rangle + 2F_0 F_1 + (F_0 - F_1)^2$, and the F -terms cancel. $\square$

Corollary 3.2 (PROVED). *If $A \geq 0$ and $B \geq 0$ pointwise on $[0, 1]^2$, then for all μ, ν, q with $M \geq m$,*

$$\text{gap}(\mu, \nu, q) \geq (1-q)^2 \langle \mu \otimes \mu, R \rangle + q^2 \langle \nu \otimes \nu, R \rangle + q(1-q)T_m \geq 0.$$

Proof. By (C1), $\text{gap} = (1-q)^2 \langle \mu \otimes \mu, D_M/M \rangle + q^2 \langle \nu \otimes \nu, D_M/M \rangle + q(1-q)[2\langle \mu \otimes \nu, D_M/M \rangle + c]$. Since $(1-q)^2, q^2, q(1-q) \geq 0$ and the measures are positive, (C4) applied pointwise gives the first inequality (with (C5) for the last bracket). $R = A + B \geq 0$ handles the first two terms, and (C6) writes T_m as a sum of pairings of A and B against positive product measures plus a square. $\square$

Remark 3.3 (what is false, and why the splitting is rank-one). On the nine-variable class, $T = a^T M_T a$ with $(M_T)_{ij} = P_2(x_i, y_j) + P_2(x_j, y_i) + Q_2(x_i, x_j) + Q_2(y_i, y_j)$ (claim C2 of the reduction artifact), so what is needed is *copositivity* of M_T . Entrywise nonnegativity is false: at $s = \tau = 0$ every Q_2 term vanishes and the entry reduces to $P_2(t, \sigma)$, with certified witness $P_2(\frac{1}{2}, \frac{1}{2}) \in [-0.055370810642371949 \pm 10^{-18}]$. Identity (C6) sidesteps copositivity: it is Diananda's decomposition of a copositive form into a nonnegative part (A on cross pairs, B on same-component pairs) plus a rank-one positive semidefinite part $(F_0 - F_1)^2$, made explicit by the choice $\varphi(s) = \sqrt{Q_2(s, s)}$. That choice saturates B on the whole diagonal ($B(s, s) = 0$) and is forced at the diagonal zeros of R , i.e. at $s \in \{0, x^*, 1\}$ (the reduction artifact's `phi_sharpness` block records slack elsewhere but no uniform rescaling that survives), which is why $B \geq 0$ had to be proved through the parameter-free Φ (Section 5).

4 $R \geq 0$ and $A \geq 0$ on the unit square

Theorem 4.1 (PROVED with MACHINE-VERIFIED strata). *At (β^*, m^*) , $R(s, t) \geq 0$ and $A(s, t) \geq 0$ for all $(s, t) \in [0, 1]^2$.*

Certificate. `liu9-h2-twovar.json`, sha256 `a028ff17ae8492211a2ef9de5721de7f3a0afd8bd45bf7d8d9b77b7c90d2e6eb`, internal digest `4ddb0f38ca881dde204f086076424617edcdbe6a2826dfa945ef963ebdf70c6`, module `uc/liu9_h2_twovar_lemmas.py` sha256 `2afe8e242aa5f506c6d54967a133fe246a7e5cb8bad4961b9860e93b86f67163`; Arb at 400 bits; the artifact also certifies $\Lambda(s, t) = 2P_2(s, t) + \beta h(\pi(s, s)) + \beta h(\pi(t, t)) \geq 0$, which is not used here.

The proof is stratified, because both functions have exact zeros: on the faces $\{s = 0\} \cup \{t = 0\}$, at the corner $(1, 1)$, and at the interior point (x^*, x^*) (Lemma 2.4). A function that touches zero cannot be certified positive by interval subdivision alone, so every zero is handled analytically and the certified cover runs only on the compact remainder. Let $F \in \{R, A\}$ and $c_0 = 1 - \beta - \frac{1}{2m}$, $c_1 = 1 - \frac{1}{2m}$; certified $c_0 \in [0.08995657486712521478 \pm 10^{-20}]$ and $c_1 \in [0.19000913473001832145 \pm 10^{-20}]$.

Lemma 4.2 (faces; PROVED). *$F(0, t) = F(s, 0) = 0$; $R(1, t) = R(t, 1) = c_1 h(t) \geq 0$ and $A(1, t) = A(t, 1) = c_0 h(t) \geq 0$.*

Proof. $h(0) = 0$, $\pi(0, t) = 0$ and $\varphi(0) = 0$ give the zero faces. $\pi(1, t) = t$ and $h(1) = 0$ give $R(1, t) = (1 - \beta)h(t) + \beta h(t) - h(t)/(2m)$; $\varphi(1) = \sqrt{\beta h(\pi(1, 1))} = \sqrt{\beta h(1)} = 0$ gives $A(1, t) = P_2(1, t) = (1 - \beta)h(t) - h(t)/(2m)$. $\square$

Lemma 4.3 (near-zero strip; PROVED). *For $0 < u < 1$, $u \log(1/u) \leq h(u) \leq u \log(1/u) + u$. Consequently*

$$F(s, t) \geq P_2(s, t) \geq st \left[c_0 \log \frac{1}{st} - \frac{1}{m} \right] \quad (0 < s, t \leq 1),$$

and if $\min(s, t) \leq \varepsilon = 10^{-8}$ then $F(s, t) \geq st [c_0 \log(1/\varepsilon) - 1/m] \geq 0$, the bracket being certified $\geq 0.03707961590679940650$.

Proof. The two-sided bound is $0 \leq -(1 - u) \log(1 - u) \leq u$. Both $Q_2 = \beta K$ and $\varphi \otimes \varphi$ are nonnegative, so $F \geq P_2$. Apply the lower bound to $h(st)$ and the upper bound to $h(s), h(t)$: $th(s) + sh(t) \leq st(\log \frac{1}{st} + 2)$, whence $P_2 \geq (1 - \beta)st \log \frac{1}{st} - \frac{st}{2m}(\log \frac{1}{st} + 2)$. If $\min(s, t) \leq \varepsilon$ then $st \leq \varepsilon$ and, as $c_0 > 0$, $c_0 \log \frac{1}{st} \geq c_0 \log \frac{1}{\varepsilon}$. $\square$

Lemma 4.4 (corner (1, 1); PROVED). *Let $a = 1 - s$, $b = 1 - t$, $u = a + b \in (0, 2\delta]$. Then, with $\mathcal{A} = a \log \frac{1}{a} + b \log \frac{1}{b} \leq u(\log \frac{1}{u} + \log 2)$,*

$$R \geq u \left[((1 - \beta)(1 - \frac{\delta}{2}) + \beta(1 - \delta) - \frac{1}{2m}) \log \frac{1}{u} - \frac{\log 2 + 1}{2m} \right], \quad A \geq u \left[((1 - \beta)(1 - \frac{\delta}{2}) - \frac{1}{2m}) \log \frac{1}{u} - \frac{\log 2 + 1}{2m} \right].$$

With $\delta = 10^{-4}$ for R and $\delta = 10^{-8}$ for A the brackets are certified $\geq 0.24644229080956936883$ and $\geq 0.22327437062436302916$ respectively on $u \in (0, 2\delta]$, so $F \geq 0$ there; $u = 0$ is the exact corner zero.

Proof. $W := 1 - st = u - ab$ and $V := 1 - \pi(s, t) = W - abst$ satisfy $ab \leq u^2/4 \leq u\delta/2$, hence $u(1 - \delta/2) \leq W \leq u$ and $u(1 - \delta) \leq V \leq u$. By the symmetry $h(z) = h(1 - z)$ and the lower bound of Lemma 4.3, $h(st) = h(W) \geq W \log \frac{1}{W} \geq u(1 - \frac{\delta}{2}) \log \frac{1}{u}$ and $K(s, t) = h(V) \geq u(1 - \delta) \log \frac{1}{u}$. The subtracted term is $th(s) + sh(t) = (1 - b)h(a) + (1 - a)h(b) \leq h(a) + h(b) \leq \mathcal{A} + u$ by the upper bound of Lemma 4.3, and $\mathcal{A} = u \log \frac{1}{u} + u H_e(a/u) \leq u(\log \frac{1}{u} + \log 2)$ with H_e the binary entropy. For A drop $\varphi \otimes \varphi \geq 0$ and use only the $h(st)$ term. $\square$

Lemma 4.5 (the interior zero; PROVED, constants MACHINE-VERIFIED). *Let $z^* = (x^*, x^*)$, let $r_0 > 0$, and suppose that on the box $\{z : |z - z^*|_\infty \leq r_0\}$ all third partial derivatives of F are bounded in absolute value by T_3 , and that $\lambda_{\min}(\nabla^2 F(z^*)) \geq \lambda > 0$. Then with $C_3 = 2\sqrt{2}T_3$ and $\rho = 3\lambda/C_3$, $F(z^* + d) \geq \frac{\lambda}{2}|d|^2 - \frac{C_3}{6}|d|^3 \geq 0$ for all d with $|d|_2 \leq \rho$ and $|d|_\infty \leq r_0$. Certified data: for R , $r_0 = 0.03$, $\lambda = 0.25681809294727316048$, $C_3 \leq 28.567618523131339446$, $\rho \geq 0.026969496187369588069$; for A , $r_0 = 0.025$, the same λ , $C_3 \leq 32.259197768814229185$, $\rho \geq 0.023883243605848030197$. The squares $[x^* - w, x^* + w]^2$ with $w = 9/500$ (R) and $w = 2/125$ (A) satisfy $\sqrt{2}(w + \eta) \leq \rho$ (hence also $w + \eta \leq r_0$), where $\eta \leq 1.19 \times 10^{-69}$ is the distance from x^* to the rational centre used by the cover; hence $F \geq 0$ on those squares.*

Proof. $F(z^*) = 0$ and $\nabla F(z^*) = 0$ by Lemma 2.4 (exact algebra; the Arb enclosures of the value and of the gradient components, of radius below 5×10^{-68} and 10^{-68} respectively, contain 0 and serve as consistency checks). Taylor's theorem with Lagrange remainder along the segment from z^* to $z^* + d$ gives $F(z^* + d) = \frac{1}{2}d^T \nabla^2 F(z^*)d + \frac{1}{6} \sum_{ijk} F_{ijk}(\xi) d_i d_j d_k$ with $|\sum_{ijk} F_{ijk}(\xi) d_i d_j d_k| \leq T_3(|d_1| + |d_2|)^3 \leq 2\sqrt{2}T_3|d|^3$. The quadratic form is $\geq \lambda|d|^2$, and $\frac{\lambda}{2}|d|^2 - \frac{C_3}{6}|d|^3 \geq 0$ iff $|d| \leq 3\lambda/C_3$. λ is bounded below by $\min(F_{ss} - |F_{st}|, F_{tt} - |F_{st}|)$ evaluated on the artifact's unrounded Hessian enclosures of F (radius $\sim 10^{-51}$; Lemma 2.4 prints them rounded, which would only give $\lambda \geq 0.25681809294727316046$); T_3 is the upper endpoint of the Arb enclosure of the third partials over the box; φ is smooth near z^* because $K(s, s) > 0$ for $0 < s < 1$. $\square$

Proof of Theorem 4.1. Lemma 4.3 covers $\{\min(s, t) \leq \varepsilon\}$ (and the zero faces), Lemma 4.4 the corner square $\{a + b \leq 2\delta\} \supseteq [1 - \delta, 1]^2$, and Lemma 4.5 the local square around z^* . For each F the remaining set, $[\varepsilon, 1]^2$ minus the local and corner squares, is the union of 14 exact rational rectangles (the 4×4 grid generated by the cut coordinates, minus the two removed cells). On each rectangle a deterministic Arb branch-and-bound certifies $F > 0$: a cell is accepted when one of three certified lower bounds is positive — the analytic product bound of Lemma 4.3, the lower endpoint of the direct Arb range of F on the cell (with the range of h taken from endpoint values and $\log 2$ exactly when the argument interval crosses $\frac{1}{2}$), or the centred first-order bound $F(\text{mid}) - \sup |F_s| w_s - \sup |F_t| w_t$ with derivative enclosures over the whole cell — and is bisected along its longer side otherwise. The run terminates with every cell accepted: for R , 7,462 cells processed, 3,738 accepted, maximal depth 43, worst certified lower bound at least 1.01×10^{-18} (on a cell adjacent to the ε -strip); for A , 16,928 processed, 8,471 accepted, depth 48. Three (for R) and four (for A) mutations of the certificate — dropping the local stratum, dropping the corner stratum, understating C_3 by a factor 16, flipping the sign of the logarithmic term in Lemma 4.3 — each make the run fail, as required. $\square$

5 $\Phi \geq 0$ on the unit square, hence $B \geq 0$

Write $\Lambda(u) = h(u)/u = \log \frac{1}{u} + \mu(u)$ for $0 < u \leq 1$, with $\mu(u) = -(1-u) \log(1-u)/u$, $\mu(0) = 1$, $\mu(1) = 0$; $\pi_d(x) = \pi(x, x)$; $g = \Lambda \circ \pi_d$ and $\gamma = \log g$ on $(0, 1)$; subscripts $\pi_{st} = \pi(s, t)$, $\Lambda_{ss} = \Lambda(\pi_{ss})$, etc.; $a = 1 - s$, $b = 1 - t$. Lemmas 5.2 and 5.3 are statements on the open square $0 < s, t < 1$, where every π argument lies in $(0, 1]$ and Λ is finite; the closed square is handled by Lemma 5.4.

Lemma 5.1 (elementary facts; PROVED).

- (a) $u h'(u) - h(u) = \log(1 - u)$, hence $\Lambda'(u) = \log(1 - u)/u^2 < 0$ on $(0, 1)$.
- (b) $\pi'_d(x) = 2x(2 - 3x + 2x^2) > 0$ on $(0, 1]$ (discriminant -7) and $\pi''_d(x) = 4(1 - 3x + 3x^2) > 0$ (discriminant -3): π_d and π'_d are increasing.
- (c) g and γ are strictly decreasing on $(0, 1)$, and $\gamma'(x) = \log(1 - \pi_d(x)) \pi'_d(x) / (\pi_d(x) h(\pi_d(x)))$.
- (d) $\mu(u) = 1 - \sum_{k \geq 2} u^{k-1} / (k(k-1))$ on $[0, 1]$: μ is decreasing and concave, $0 \leq \mu \leq 1$, continuous on $[0, 1]$.
- (e) $\pi_{ss}\pi_{tt} - \pi_{st}^2 = s^2 t^2 (s - t)^2$.
- (f) $M_\mu := \mu(\pi_{ss}) + \mu(\pi_{tt}) - 2\mu(\pi_{st}) \leq 0$.
- (g) $\Lambda \geq 0$ on $(0, 1]$.

Proof. (a) $h'(u) = \log \frac{1-u}{u}$, so $uh' - h = u \log(1-u) - u \log u + u \log u + (1-u) \log(1-u)$. (b) is a direct computation. (c) chain rule with (a), (b), and $\Lambda(\pi_d) = h(\pi_d)/\pi_d$. (d) $-(1-u) \log(1-u) = u - \sum_{k \geq 2} u^k / (k(k-1))$ from the series of $-\log(1-u)$; all coefficients after the constant are negative. (e) $\pi_{ss} = s^2(1+a^2)$, $\pi_{tt} = t^2(1+b^2)$, $\pi_{st} = st(1+ab)$, and $(1+a^2)(1+b^2) - (1+ab)^2 = (a-b)^2 = (s-t)^2$. (f) by (e), $\pi_{st} \leq \sqrt{\pi_{ss}\pi_{tt}} \leq \frac{1}{2}(\pi_{ss} + \pi_{tt})$; μ decreasing gives $\mu(\pi_{st}) \geq \mu(\frac{1}{2}(\pi_{ss} + \pi_{tt}))$ and concavity gives $\mu(\frac{1}{2}(\pi_{ss} + \pi_{tt})) \geq \frac{1}{2}(\mu(\pi_{ss}) + \mu(\pi_{tt}))$. (g) $h \geq 0$. $\square$

Lemma 5.2 (exact identity; PROVED). For $0 < s, t < 1$, with $\delta = (s - t)^2 / (1 + ab)^2$ (so $\pi_{ss}\pi_{tt} = \pi_{st}^2(1 + \delta)$) and $\bar{\Lambda} = \frac{1}{2}(\Lambda_{ss} + \Lambda_{tt})$,

$$\Phi = \pi_{st}^2 \left[\frac{1}{4}(\Lambda_{ss} - \Lambda_{tt})^2 + \frac{1}{2}(\log(1 + \delta) - M_\mu)(\Lambda_{st} + \bar{\Lambda}) \right] - s^2 t^2 (s - t)^2 \Lambda_{ss} \Lambda_{tt},$$

and therefore, since the middle term is nonnegative,

$$\Phi \geq s^2 t^2 \left[\frac{1}{4}(1 + ab)^2 (g(s) - g(t))^2 - (s - t)^2 g(s) g(t) \right]. \quad (8)$$

Proof. $K = \pi\Lambda(\pi)$, so $\Phi = \pi_{st}^2(\Lambda_{st}^2 - \Lambda_{ss}\Lambda_{tt}) - (\pi_{ss}\pi_{tt} - \pi_{st}^2)\Lambda_{ss}\Lambda_{tt}$, and $\Lambda_{st}^2 - \Lambda_{ss}\Lambda_{tt} = \frac{1}{4}(\Lambda_{ss} - \Lambda_{tt})^2 + (\Lambda_{st} - \bar{\Lambda})(\Lambda_{st} + \bar{\Lambda})$. Since $\Lambda = \log \frac{1}{\cdot} + \mu$, $2(\Lambda_{st} - \bar{\Lambda}) = \log \frac{\pi_{ss}\pi_{tt}}{\pi_{st}^2} - M_\mu = \log(1 + \delta) - M_\mu$, and Lemma 5.1(e) gives $\pi_{ss}\pi_{tt} - \pi_{st}^2 = s^2t^2(s-t)^2$ and $\delta = s^2t^2(s-t)^2/\pi_{st}^2$. The middle term is ≥ 0 by $\log(1 + \delta) \geq 0$, (f) and (g). Dropping it and writing $\pi_{st} = st(1 + ab)$, $\Lambda_{ss} = g(s)$, $\Lambda_{tt} = g(t)$ gives (8). $\square$

Lemma 5.3 (master condition; PROVED). *For $0 < s < t < 1$ put $\kappa(s, t) = (1 + ab) \frac{\gamma(s) - \gamma(t)}{t - s} > 0$. Then*

$$\Phi(s, t) \geq s^2t^2(s-t)^2g(s)g(t)\left(\frac{\kappa^2}{4} - 1\right);$$

in particular $\kappa \geq 2$ implies $\Phi \geq 0$, and $\kappa \geq \frac{9}{4}$ implies $\Phi \geq \frac{17}{64}s^2t^2(s-t)^2g(s)g(t)$.

Proof. With $x = \frac{1}{2}(\gamma(s) - \gamma(t)) > 0$, $g(s) - g(t) = e^{\gamma(s)} - e^{\gamma(t)} = 2\sqrt{g(s)g(t)} \sinh x$, so the bracket in (8) equals $(s-t)^2g(s)g(t)[((1+ab) \sinh x/(t-s))^2 - 1]$, and $\sinh x \geq x$ gives $(1+ab) \sinh x/(t-s) \geq \kappa/2$. $\square$

Lemma 5.4 (boundary and symmetry; PROVED). $\Phi(s, s) = 0$, $\Phi(0, t) = \Phi(s, 0) = 0$, $\Phi(s, 1) = \Phi(1, s) = h(s)^2 \geq 0$, and $\Phi(s, t) = \Phi(t, s)$. Hence $\Phi \geq 0$ on $[0, 1]^2$ follows from $\kappa \geq 2$ on the open triangle $0 < s < t < 1$.

Proof. $\pi(0, t) = 0$, $\pi(s, 1) = s$, $\pi(1, 1) = 1$ and $h(0) = h(1) = 0$. $\square$

Lemma 5.5 (derivative bounds; PROVED).

(N0) For $0 < x \leq x_0 < e^{-(1+\log 2)/2}$: $|\gamma'(x)| \geq \nu_0(x_0) := \frac{2 - 3x_0}{x_0(2\log(1/x_0) + 1 - \log 2)}$.

(N1) For $\frac{3}{4} \leq x_1 \leq x < 1$, with $\ell_1 = \log \frac{1}{1 - \pi_d(x_1)}$: $|\gamma'(x)| \geq \nu_1(x_1) := \frac{\pi'_d(x_1)}{1 - \pi_d(x_1)} \cdot \frac{\ell_1}{\ell_1 + 1}$.

(L5) $N(x) := -\log(1 - \pi_d(x)) \pi'_d(x)$ is positive and strictly increasing on $(0, 1)$, so for $[l, r] \subset (0, 1)$: $\inf_{[l, r]} |\gamma'| \geq N(l)/(\pi_d(r) \max_{[\pi_d(l), \pi_d(r)]} h)$, where the maximum of h over an interval is $\log 2$ if the interval contains $\frac{1}{2}$ and the larger endpoint value otherwise.

Proof. (N0) $-\log(1 - \pi) \geq \pi$; $\pi'_d(x) \geq 2x(2 - 3x_0)$ for $x \leq x_0$; $h(\pi) \leq \pi(\log \frac{1}{\pi} + 1)$ (Lemma 4.3); $\pi_d(x) \leq 2x^2$ and $u \mapsto u(\log \frac{1}{u} + 1)$ is increasing on $(0, 1)$, so $h(\pi_d(x)) \leq 2x^2(2\log \frac{1}{x} - \log 2 + 1)$; hence $|\gamma'(x)| \geq \pi'_d/h(\pi_d) \geq (2 - 3x_0)/(x(2\log \frac{1}{x} + 1 - \log 2))$, and the denominator is increasing for $x < e^{-(1+\log 2)/2}$ (its derivative is $2\log \frac{1}{x} - 1 - \log 2$), so it is at most its value at x_0 . (N1) $\pi_d \leq 1$ and $h(\pi) = h(1 - \pi) \leq (1 - \pi)(\log \frac{1}{1 - \pi} + 1)$, so $|\gamma'| \geq \frac{\ell \pi'_d}{(1 - \pi_d)(\ell + 1)}$ with $\ell = \log \frac{1}{1 - \pi_d}$; on $[\frac{3}{4}, 1]$ the factor $2 - 3x + 2x^2$ increases (vertex at $\frac{3}{4}$), so $\pi'_d \geq \pi'_d(x_1)$; $\ell/(\ell + 1)$ increases in ℓ , and ℓ and $1/(1 - \pi_d)$ increase in x . (L5) N is a product of positive increasing functions (Lemma 5.1(b)); $|\gamma'(\xi)| = N(\xi)/(\pi_d(\xi)h(\pi_d(\xi)))$ and h increases on $[0, \frac{1}{2}]$ and decreases after. $\square$

Theorem 5.6 (PROVED with a MACHINE-VERIFIED finite cover). $\Phi(s, t) \geq 0$ on $[0, 1]^2$; quantitatively, $\Phi \geq \frac{17}{64}s^2t^2(s-t)^2g(s)g(t)$ for $0 < s, t < 1$.

Certificate. liu9-h2-boundary.json, sha256 e4d6d96df3ae435800cef5739fb1ae45b98febf539ab418df116ceb03bd481c9, internal digest 389804b249cd8b16a47f3c6ab64a70ad4a9afe78097b4a8c6739759804c9eaaa, module uc/liu9_h2_boundary_layer.py sha256 fcb3ed55e8bd9ad6d2f180f5ffa79f192dacc940a5c7506eca2a85aa612905e6 (Arb, 400 bits). Independent constructive audit, own cover at $\theta = 23/10$: liu9-h2-p-hi-audit.json, sha256 f2359fa3b3b08bab29965253d777dcf32e4b1aee1e0bacfb0924b86ac163996, internal digest a515102c1807849b606bf4a681bd0a67173445b1a22610d397bb779440222cab, module uc/liu9_h2_phi_audit.py sha256 b1d9ed3d9da9e7c99c026843b683447809fb38d28dba4c49a087729bf66cb50c.

Proof. By Lemmas 5.3 and 5.4 it suffices to certify $\kappa \geq \theta := \frac{9}{4}$ on $0 < s < t < 1$. For a rational cell $C = [s_1, s_2] \times [t_1, t_2]$ with $s_1 < t_2$, every $(s, t) \in C$ with $0 < s < t < 1$ satisfies, by the mean value theorem and Lemma 5.1(c),

- (D) $\kappa \geq (1 + (1 - s_2)(1 - t_2)) \cdot \inf_{\xi \in [s_1, t_2] \cap (0, 1)} |\gamma'(\xi)|$, the infimum being bounded below by the minimum of $\nu_0(\min(t_2, x_0))$ on $(0, x_0]$, $\nu_1(\max(s_1, x_1))$ on $[x_1, 1)$, and (L5) on the middle piece, whichever apply (with $x_0 = \frac{1}{8}$, $x_1 = \frac{7}{8}$);
- (Q) if $s_2 < t_1$: $\kappa \geq (1 + (1 - s_2)(1 - t_2)) (\gamma(s_2) - \gamma(t_1)) / (t_2 - s_1)$.

Starting from the single cell $[0, 1]^2$, a cell is accepted when the lower endpoint of the Arb enclosure of (D) or (Q) exceeds θ , discarded when $s_1 \geq t_2$ (it contains no point with $s < t$), and bisected along its longer side otherwise. The run terminates: 30,974 cells processed, 792 accepted by (D), 14,687 by (Q), 17 discarded, maximal depth 23, minimum certified bound $[2.2500008364172621445 \pm 10^{-19}] > \theta$. Five mutations are caught: raising θ to $12/5$ (above the true minimum of κ ; witness $\kappa(\frac{27}{50}, \frac{27}{50} + 10^{-6}) \in [2.3941933797780204 \pm 10^{-16}] < 12/5$, so the cover cannot be tightened past the true minimum); dropping the factor $(1 + ab)$ (witness $|\gamma'(\frac{9}{20})| \in [1.9105187072037912 \pm 10^{-16}] < 2$, so the factor is load-bearing); the wrong protocol $s + t - st$; $\frac{1}{4}$ replaced by $\frac{1}{2}$ in Lemma 5.2; and (N0) applied at $x_0 = \frac{1}{2}$ beyond its validity. The audit artifact re-derives Lemma 5.2 in its own basis (residuals at most 9.4×10^{-120} in Arb and 1.3×10^{-101} in 100-digit mpmath at 22 points down to 2^{-40} from the edges and the diagonal), builds an independent cover at $\theta = 23/10$ from a 16×16 grid with its own strip lemmas at $\frac{1}{16}$ and $\frac{15}{16}$ (88,390 cells, depth 17, accepted area $2151/4096$ plus discarded $1945/4096$ equal to 1 exactly, minimum certified bound ≥ 2.30000075), reproduces both witnesses, and catches 6/6 of its own mutations. $\square$

Remark 5.7. κ diverges to $+\infty$ at every boundary point of the square (Lemma 5.5), so the singularity of h' at entropy arguments 0 and 1 — which defeats direct interval subdivision of Φ near the edges — never enters: the endpoint strips are the easy part of the cover. The numerical minimum of κ is ≈ 2.394 at $s = t \approx 0.54$ (COMPUTATIONAL-EVIDENCE), a uniform margin of 20% over the threshold 2.

Corollary 5.8 ($B \geq 0$; PROVED). *For every $\beta > 0$, $B = \beta(K_{st} - \sqrt{K_{ss}K_{tt}}) \geq 0$ on $[0, 1]^2$.*

Proof. $\Phi = (K_{st} - \sqrt{K_{ss}K_{tt}})(K_{st} + \sqrt{K_{ss}K_{tt}})$ with $K \geq 0$. If the second factor is positive, $B = \beta\Phi/(K_{st} + \sqrt{K_{ss}K_{tt}}) \geq 0$; if it vanishes then $K_{st} = 0 = K_{ss}K_{tt}$ and $B = 0$. $\square$

6 Hypothesis 2 for all Borel probability measures

Theorem 6.1 (PROVED, conditional only on the cited artifacts). *Theorem 1.3 holds: for all Borel probability measures μ, ν on $[0, 1]$ and all $q \in [0, 1]$, $\text{gap}(\mu, \nu, q) \geq \frac{M-m^*}{m^*} \langle P, h \rangle$, and in particular $\text{gap} \geq 0$ when $M \geq m^*$.*

Certificate. liu9-h2-general-lift.json, sha256 eb4874d39904593bddc3f2ae638d1bad6756a19d4ebf6e06a60427b6bb7ac1ce, internal digest 3ff1e797cd5dbcc4273d9f002b9e540b58829d237b2d344f266e0bea46c02bc8, module uc/liu9_h2_general_lift.py sha256 b265e726d110098a09c00de331c4433454973ba741966b3018b77afcc225b5c3 (Arb 400 bits and sympy; seven mutations caught; pins the artifacts of Sections 3–5 by hash and claim status and checks that they use the same definitions of the constants).

Proof. The statement with $\text{gap} \geq 0$ for $M \geq m^*$ is Corollary 3.2 with Theorem 4.1 ($A \geq 0$, $R \geq 0$) and Corollary 5.8 ($B \geq 0$). The quantitative form follows from Theorem 7.1 below (the two-component case), or directly: by (C4), $\text{gap} - \frac{M-m}{m} \langle P, h \rangle$ equals the expression obtained from (4) with M replaced by m in the bracket, i.e. the right-hand side of Corollary 3.2, which is ≥ 0 . Lemma 2.1 guarantees that all pairings are finite and that integrating a pointwise inequality

between continuous kernels against a positive product measure preserves it; no density or continuity argument in the measures is needed, so the fact that h is not Lipschitz at $0, 1$ is irrelevant. $\square$

Remark 6.2 (Liu’s nine-parameter form; product coupling; witnesses). Liu’s (81)–(84) is $\text{gap} \geq 0$ restricted to $\mu = \sum_{i=1}^3 a_i \delta_{x_i}$, $\nu = \sum_{i=1}^3 a_i \delta_{y_i}$ (shared masses) with $M \geq 1 - c$; Hypothesis 2 at $c = c'$ is the statement that the minimum of the ratio $\text{num}/\langle P, h \rangle$ over this class is ≥ 1 , and Theorem 6.1 gives $\text{ratio} \geq M/m^* \geq 1$ for every μ, ν with $\langle P, h \rangle > 0$ (when $\langle P, h \rangle = 0$, i.e. P is supported on $\{0, 1\}$, the numerator vanishes too and the cross-multiplied inequality reads $0 \geq 0$). Conversely, any finitely supported pair $\mu = \sum_i a_i \delta_{x_i}$, $\nu = \sum_j b_j \delta_{y_j}$ is the paired configuration with shared masses $g_{ij} = a_i b_j$ and supports (x_i, y_j) , because gap depends on the configuration only through its marginals (a polynomial identity in the masses, checked by sympy at seven sizes in the artifact). The artifact records four witnesses at the exact balls: W1a, $\text{gap} = 0$ at Liu’s minimiser (equality by the algebra of Lemma 2.4: $D_{m^*}(0, \cdot) = 0$ and $D_{m^*}(x^*, x^*) = m^* h(x^{*2}) - x^* h(x^*) = 0$); W1b, the certified negative gap at the truncated constants of Remark 2.5; and W2, W3, certified negative gaps ($\text{gap} \leq -0.1249$ at $\mu = \nu = \delta_{1/2}$ and $\text{gap} \leq -0.1344$ at a two-atom/one-atom pair with mean $43/90$) showing that the mean constraint $M \geq m^*$ is load-bearing. A sample of 100 certified mean-feasible random pairs (minimum certified $\text{gap} \geq 0.0058665$) is recorded as COMPUTATIONAL-EVIDENCE only.

7 The mixture theorem

Theorem 7.1 (PROVED, identities MACHINE-VERIFIED). *Let $\Gamma = \int \nu_u \otimes \nu_u dP(u)$ be a conditionally i.i.d. coupling of $\mu = \int \nu_u dP(u)$, and let $M = \langle \mu, x \rangle$. At (β^*, m^*) ,*

$$\begin{aligned} \text{num}(\Gamma) - \langle \mu, h \rangle &= \iint \langle \nu_u \otimes \nu_v, A \rangle dP(u) dP(v) + \int \langle \nu_u \otimes \nu_u, B \rangle dP(u) \\ &\quad + \text{Var}_P(\langle \nu_U, \varphi \rangle) + \frac{M - m^*}{m^*} \langle \mu, h \rangle, \end{aligned}$$

and consequently $\text{num}(\Gamma) \geq \frac{M}{m^*} \langle \mu, h \rangle$.

Certificate. liu9-h2-mixture-theorem.json, sha256 329f7e2d71af8cd78d1a921c72b9ae4d05b71113134eb3341d69932f19ea24b3, internal digest b108b781224dff601ffcf3f562703d3b620e7da6d89cf6d000b6ccbd06a4773b, module uc/liu9_h2_mixture_theorem.py sha256 3098a1ca30a0f16582df02e1fb7dfd8fa27970c031167125e696e58ac43dfbab: the identities (I)–(III) below as exact rational-function identities in the free pairing algebra for 2, 3, 4 components with symbolic weights (sympy residual 0), in Arb to 10^{-100} at 60 random mixtures with up to 6 components and up to 4 atoms each (atoms at 0 and 1 included), agreement with Liu’s audited transcription at two components to 10^{-100} , six mutations caught.

Proof. Write $K_{uv} = \langle \nu_u \otimes \nu_v, K \rangle$, $P_{uv} = \langle \nu_u \otimes \nu_v, P_2 \rangle$, $F_u = \langle \nu_u, \varphi \rangle$, $H = \langle \mu, h \rangle$. All integrands below are bounded and measurable ($u \mapsto \nu_u$ is a Markov kernel and every kernel is bounded continuous, Lemma 2.1), so Fubini applies, and by bilinearity $\langle \mu \otimes \mu, k \rangle = \iint \langle \nu_u \otimes \nu_v, k \rangle dP dP$ for every kernel k . Three identities:

$$\begin{aligned} \text{(I)} \quad & \int K_{uu} dP - \iint K_{uv} dP dP = \frac{1}{2} \iint (K_{uu} + K_{vv} - 2K_{uv}) dP dP, \\ \text{(II)} \quad & \langle \mu \otimes \mu, (1 - \beta)h(xy) + \beta K \rangle - H = \langle \mu \otimes \mu, R \rangle + \frac{M - m}{m} H, \\ \text{(III)} \quad & \langle \mu \otimes \mu, R \rangle + \frac{\beta}{2} \iint (K_{uu} + K_{vv} - 2K_{uv}) dP dP \\ &= \iint \langle \nu_u \otimes \nu_v, A \rangle dP dP + \int \langle \nu_u \otimes \nu_u, B \rangle dP + \text{Var}_P(F_U). \end{aligned}$$

(I) uses only $\int dP = 1$. (II): $R = (1 - \beta)h(xy) + \beta K - \frac{y h(x) + x h(y)}{2m}$ and, by the product rule, $\langle \mu \otimes \mu, y h(x) + x h(y) \rangle = 2MH$. (III): the left side is $\iint (P_{uv} + \beta K_{uv}) dP dP + \beta \int K_{uu} dP - \beta \iint K_{uv} dP dP = \iint P_{uv} dP dP + \beta \int K_{uu} dP$; the right side is $\iint P_{uv} dP dP + (\int F_u dP)^2 + \beta \int K_{uu} dP - \int F_u^2 dP + \int F_u^2 dP - (\int F_u dP)^2$, by the product rule for $\varphi \otimes \varphi$. Now $\text{num}(\Gamma) - H = \langle \mu \otimes \mu, (1 - \beta)h(xy) + \beta K \rangle - H + \beta [\int K_{uu} dP - \iint K_{uv} dP dP]$, and (I), (II), (III) in turn give the displayed identity. The first two terms are ≥ 0 by Theorem 4.1 and Corollary 5.8 integrated against positive product measures, and the third is a variance. $R \geq 0$ is not used. $\square$

Remark 7.2 (two components; the paired chain). For $P = (1 - q)\delta_0 + q\delta_1$, $\nu_0 = \mu$, $\nu_1 = \nu$, the variance is $q(1 - q)(F_0 - F_1)^2$ and the channel $\frac{\beta}{2} \iint (\cdots)$ is $q(1 - q)c$; the displayed identity is then (C1)+(C4)+(C6) of Theorem 3.1, and Theorem 6.1 is the special case.

Lemma 7.3 (sharpness; PROVED). *For $\mu_w = (1 - w)\delta_0 + w\delta_{x^*}$, $0 \leq w \leq 1$, and the one-component coupling $\Gamma = \mu_w \otimes \mu_w$: $\text{num}(\Gamma) = \frac{M}{m^*} \langle \mu_w, h \rangle$ exactly. Laws supported on $\{0, 1\}$ give $\text{num} = \langle \mu, h \rangle = 0$. Hence the constant M/m^* in Theorem 7.1 is attained at every mean $M \in [0, x^*]$, and $c' = 1 - m^*$ is the largest constant obtainable from this two-protocol inequality at β^* .*

Proof. Cross terms vanish because $h(0 \cdot x) = 0$ and $K(0, x) = h(0) = 0$, so $\text{num} = w^2[(1 - \beta)h(x^{*2}) + \beta K(x^*, x^*)] = w^2 h(x^{*2})$ by Lemma 2.3(b). Also $\langle \mu_w, h \rangle = w h(x^*)$ and $M = wx^*$, so $\frac{M}{m^*} \langle \mu_w, h \rangle = w^2 \frac{x^* h(x^*)}{m^*} = w^2 h(x^{*2})$. For laws on $\{0, 1\}$ every entropy vanishes. Arb enclosures of the residual at the exact balls contain 0 at $w = \frac{3}{10}, \frac{1}{2}, \frac{9}{10}, 1$. $\square$

Remark 7.4. Liu's minimiser is $w = p^*$ ($M = m^*$); Lemma 7.3 shows the equality is not a property of that single point but of the ray through δ_{x^*} . The artifact also certifies a five-component mixture with $M > m^*$ whose ratio exceeds M/m^* strictly, and, by Remark 2.5, that at the 100-digit truncations the bound is false at $w = p^*$ (certified -5.95×10^{-102}).

8 From union-closed families to the constant

This section re-derives Liu's Proposition 3 from his Definition 1 and Example 5 so that the dependence on his paper is limited to definitions. Every step is elementary; it is labelled PROVED (human argument, re-derived independently by two readers, cf. Section 10). Entropies may be taken in any base: the accounting below is base-free, and the inequality of Theorem 7.1 is homogeneous in the base.

Setting. Identify subsets of $E = \{1, \dots, n\}$ with 0/1 vectors and let $X^n = (X_1, \dots, X_n)$ be uniform on $\mathcal{F}$, so $H(X^n) = \log |\mathcal{F}|$. For $i \geq 1$ and a past x^{i-1} of positive probability write $\sigma_i(x^{i-1}) = \mathbb{P}(X_i = 1 \mid X^{i-1} = x^{i-1})$.

Definition 8.1 (protocol; Liu, Definition 1). A protocol Π assigns to each $(s, t) \in [0, 1]^2$ a law $\Pi_{s,t}$ on $\{0, 1\}^2$ with marginal means s and t . Given Π , generate $(\tilde{X}^n, \tilde{Y}^n)$ sequentially: at step i , with $S_i = \sigma_i(\tilde{X}^{i-1})$ and $T_i = \sigma_i(\tilde{Y}^{i-1})$, draw $(\tilde{X}_i, \tilde{Y}_i) \sim \Pi_{S_i, T_i}$ using randomness independent of everything before.

Lemma 8.2 (PROVED). *For every protocol: (i) $\tilde{X}^n \sim X^n$ and $\tilde{Y}^n \sim X^n$, and only pasts of positive probability are visited; (ii) the law of S_i is the law μ_i of $\sigma_i(X^{i-1})$, with $\mathbb{E} S_i = \mathbb{P}(X_i = 1)$; (iii)*

$$H(\tilde{X}^n \vee \tilde{Y}^n) \geq \sum_{i=1}^n \mathbb{E}[h(\Pi_{S_i, T_i}(0, 0))], \quad H(X^n) = \sum_{i=1}^n \mathbb{E}[h(S_i)],$$

where $\vee$ is the coordinatewise maximum; and (iv) if $\mathcal{F}$ is union-closed then $H(\tilde{X}^n \vee \tilde{Y}^n) \leq H(X^n)$.

Proof. (i) $\mathbb{P}(\tilde{X}_i = 1 \mid \tilde{X}^{i-1}, \tilde{Y}^{i-1}) = S_i = \sigma_i(\tilde{X}^{i-1})$ by the marginal condition, so by induction on i the law of $\tilde{X}^i$ is that of X^i ; likewise for $\tilde{Y}$. (ii) follows from (i), and $\mathbb{E}S_i = \mathbb{E}\mathbb{P}(X_i = 1 \mid X^{i-1}) = \mathbb{P}(X_i = 1)$. (iii) By the chain rule and because $(\tilde{X} \vee \tilde{Y})^{i-1}$ is a function of $(\tilde{X}^{i-1}, \tilde{Y}^{i-1})$, $H(\tilde{X}^n \vee \tilde{Y}^n) = \sum_i H((\tilde{X} \vee \tilde{Y})_i \mid (\tilde{X} \vee \tilde{Y})^{i-1}) \geq \sum_i H((\tilde{X} \vee \tilde{Y})_i \mid \tilde{X}^{i-1}, \tilde{Y}^{i-1})$; conditionally on the pasts, $(\tilde{X}_i, \tilde{Y}_i) \sim \Pi_{S_i, T_i}$, so $(\tilde{X} \vee \tilde{Y})_i$ is Bernoulli with $\mathbb{P}(= 0 \mid \text{pasts}) = \Pi_{S_i, T_i}(0, 0)$ and the conditional entropy is $\mathbb{E}[h(\Pi_{S_i, T_i}(0, 0))]$, using $h(u) = h(1 - u)$. The second formula is the chain rule for X^n with (ii). (iv) $\tilde{X}^n \vee \tilde{Y}^n$ is the indicator of the union of two members of $\mathcal{F}$, hence takes values in $\mathcal{F}$, and the uniform law maximises entropy on a given support. $\square$

The two protocols.

- (i) Gilmer's protocol $\Pi_{s,t}^{(1)} = \text{Ber}(s) \otimes \text{Ber}(t)$, $\Pi_{s,t}^{(1)}(0, 0) = (1 - s)(1 - t)$. The two coordinates use independent randomness at every step, so $\tilde{X}^{i-1}$ and $\tilde{Y}^{i-1}$ are independent and $(S_i, T_i) \sim \mu_i \otimes \mu_i$.
- (ii) Liu's Example 5 with $f(x) = x(1 - x)$. Let U be uniform on $[0, 1]$ and $\sigma(U) = +1$ if $U > \frac{1}{2}$, -1 otherwise. Given (s, t) and U , set $\tilde{X} = 0$ with probability $(1 - s) + f(1 - s)\sigma(U) = (1 - s)(1 + s\sigma(U))$ using $\tilde{X}$'s local randomness, and $\tilde{Y} = 0$ with probability $(1 - t)(1 + t\sigma(U))$ using $\tilde{Y}$'s local randomness and the same U . These are probabilities ($0 \leq (1 - s)(1 \pm s) \leq 1$), the means are s and t ($\mathbb{E}\sigma(U) = 0$), and

$$\Pi_{s,t}^{(2)}(0, 0) = \mathbb{E}[(1 - s)(1 + s\sigma)(1 - t)(1 + t\sigma)] = (1 - s)(1 - t)(1 + st) = \pi(1 - s, 1 - t),$$

since $\mathbb{E}\sigma = 0$ and $\sigma^2 = 1$. In the sequential construction a fresh U_i is drawn at step i and shared by the two coordinates. Each $\tilde{X}_j$ is the same measurable function G_j of $(\tilde{X}^{j-1}, U_j, \xi_j)$ as $\tilde{Y}_j$ is of $(\tilde{Y}^{j-1}, U_j, \eta_j)$, where (ξ_j) , (η_j) are the local randomness and (ξ_j) , (η_j) , (U_j) are independent. Hence, conditionally on U^{i-1} , $\tilde{X}^{i-1}$ and $\tilde{Y}^{i-1}$ are i.i.d., and so are $S_i = \sigma_i(\tilde{X}^{i-1})$ and $T_i = \sigma_i(\tilde{Y}^{i-1})$: the law of (S_i, T_i) is $\Gamma_i = \int \nu_u \otimes \nu_u dP_{U^{i-1}}(u)$, with ν_u the regular conditional law of S_i given $U^{i-1} = u$ and barycentre μ_i . This is Liu's observation after his Example 5.

Theorem 8.3 (PROVED). *Theorem 1.4 holds.*

Proof. If $|\mathcal{F}| = 1$ then $\mathcal{F} = \{A\}$ with $A \neq \emptyset$ and every element of A has frequency 1. Let $|\mathcal{F}| \geq 2$, so $H(X^n) > 0$. Fix $c < c'$ and suppose, for contradiction, that $\mathbb{P}(X_i = 1) < c$ for every i . Let $\bar{\mu}_i$, $\bar{\Gamma}_i$, $\bar{\nu}_u$ denote the images of μ_i , Γ_i , ν_u under $s \mapsto 1 - s$; then $\bar{\Gamma}_i = \int \bar{\nu}_u \otimes \bar{\nu}_u dP_{U^{i-1}}$ is a conditionally i.i.d. coupling of $\bar{\mu}_i$, whose mean is $M_i = \mathbb{E}[1 - S_i] = 1 - \mathbb{P}(X_i = 1) > 1 - c$, and $\langle \bar{\mu}_i, h \rangle = \mathbb{E}[h(S_i)]$ by the symmetry of h . By Lemma 8.2(iii) for the two protocols,

$$\begin{aligned} & (1 - \beta^*) H(\tilde{X}^{(1)n} \vee \tilde{Y}^{(1)n}) + \beta^* H(\tilde{X}^{(2)n} \vee \tilde{Y}^{(2)n}) \\ & \geq \sum_i \left[(1 - \beta^*) \langle \bar{\mu}_i \otimes \bar{\mu}_i, h(xy) \rangle + \beta^* \int \langle \bar{\nu}_u \otimes \bar{\nu}_u, K \rangle dP_{U^{i-1}} \right] = \sum_i \text{num}(\bar{\Gamma}_i) \\ & \geq \sum_i \frac{M_i}{m^*} \mathbb{E}[h(S_i)] \geq \frac{1 - c}{m^*} \sum_i \mathbb{E}[h(S_i)] = \frac{1 - c}{m^*} H(X^n), \end{aligned}$$

by Theorem 7.1 for each i and $\mathbb{E}[h(S_i)] \geq 0$. By Lemma 8.2(iv) the left side is at most $(1 - \beta^*)H(X^n) + \beta^*H(X^n) = H(X^n)$. Since $H(X^n) > 0$ this forces $(1 - c)/m^* \leq 1$, i.e. $c \geq 1 - m^* = c'$, contradicting $c < c'$. Hence for every $c < c'$ some i has $\mathbb{P}(X_i = 1) \geq c$; as the n frequencies are fixed numbers, $\max_i \mathbb{P}(X_i = 1) \geq c'$, i.e. some $i \in E$ lies in at least $c'|\mathcal{F}|$ members. $\square$

Remark 8.4 (on Liu’s argument, and what is not used). Liu’s Proposition 3 asks, for every $c < c'$, for a constant $C > 1$ in $\sum_k w_k \inf_{P_{ST} \in \mathcal{C}_k(\mu)} \mathbb{E}[h(\Pi_{S,T}^{(k)}(0, 0))] \geq C \mathbb{E}[h(S)]$ over all laws μ with mean $\leq c$; Theorem 7.1 gives it with the explicit $C = (1 - c)/m^*$, which exceeds 1 precisely when $c < c'$, so no limiting or separate strictness argument is needed. The couplings that actually occur are the mixtures $\bar{\Gamma}_i$, so Theorem 7.1 is applied to them directly; Liu’s Theorem 9 (reduction to two-point mixing laws), Lemma 11 (the positive-semidefiniteness of Section V-A, “Hypothesis 1”) and Theorem 12 (the Krein–Milman reduction to three shared-mass atoms) are not invoked, and neither is the weak closure $\mathcal{C}_3(\mu)$. Section V-B’s assertion that the nine-parameter optimum is ≥ 1 is replaced by Theorem 6.1. Laws supported on $\{0, 1\}$ (his Lemma 7) need no separate treatment: they contribute $0 \geq 0$.

9 The value of c'

Theorem 9.1 (MACHINE-VERIFIED).

$$c' = 1 - m^* \in [0.382709087918735029930312902098972611626381433 \pm 4.95 \times 10^{-46}];$$

in particular $c' > 0.38270908791873$.

Certificate. `liu9-h2-mixture-theorem.json` and `liu9-h2-general-lift.json` (Table 1), `constants.exact_ba`.

The enclosure is propagated in Arb from the certified rational bracket of x^* (Lemma 2.3) through the closed forms of Definition 2.2, in the artifacts of Sections 6 and 7, and agrees with the independent 100-digit solver `uc/liu9_binding.py` (difference enclosures containing 0). It exceeds $c^* \approx 0.3823455$ of [8, 2] by 3.6×10^{-4} .

Liu’s printed values (91)–(94) read $p^* \approx 0.893604513905457$, $x^* \approx 0.690787593924988$, $c' \approx 0.382709087918741$, $\beta^* \approx 0.100052559862974$. Of these, (92) agrees with the certified enclosure to all printed digits, while for (91), (93) and (94) the printed-minus-certified differences are approximately -8.4×10^{-15} , $+6.0 \times 10^{-15}$ and $+8.1 \times 10^{-14}$ respectively, i.e. they are wrong in their last two digits (last three for β^*); the correct 15-digit roundings are $p^* \approx 0.893604513905465$, $c' \approx 0.382709087918735$, $\beta^* \approx 0.100052559862893$. The constants are defined by the same equations (87)–(90), so the discrepancy is in the printed decimals only and does not affect any statement above.

10 Verification: artifacts, replay, labels, scope

Artifacts. Table 1 lists the seven machine-checked artifacts, all under `math/uc/verification/results/`, produced by the modules under `math/uc/`. Each artifact is a JSON document with a `claim_status`, an internal digest `report_sha256` (the SHA-256 of the canonical compact serialisation of the document with that field omitted or blanked, in the module’s declared scope), and, where the module records it, the SHA-256 of the module source that produced it. The artifacts of Sections 6 and 7 pin the file hashes and claim statuses of the artifacts they depend on and fail if they change; the audit artifact pins the certificate it audits.

Replay. From `math/`,

```
nice -n 19 ./venv/bin/python -I -B uc/verification/replay_h2_chain.py
```


artifact (module)	claim; status
liu9-h2-reduction.json (liu9_h2_reduction.py) file c63ae94e4fbd2397e19ea9e2a4e2c155f1e02265852d0aacad5c97d70f078e3a internal c40d27be152217bd0d862bf54c4a957ab7bcad22e6b2f7747cf7462e9bc7a397 module 99840c239eb60c3db3a1093827f3c9849c5d010debec70eb13661a87e0418fe0	(C1)–(C6), Theorem 3.1; REDUCTION-CERTIFIED
liu9-psi-reduction.json (liu9_psi_reduction_audit.py) file 9fb5766f1c3110cce9b231d2f2f1b7c1eec67bc3d6ac7602be50fc72b00c07a3 internal 405eea478f2146fde762f749dd4a8f6986be79f0be47f56cb2cc24daf64b2af0 module 18602dc7c8599ab6752583535ed5809df571deb8d8851e4accb4d07362e927bb	independent re-derivation of the chain; PROVED
liu9-h2-twovar.json (liu9_h2_twovar_lemmas.py) file a028ff17ae8492211a2ef9de5721de7f3a0afd8bd45bf7d8d9b77b7c90d2e6eb internal 4ddb0f38ca881dde204f086076424617edcbe6a2826dfa945ef963ebdf70c6 module 2afe8e242aa5f506c6d54967a133fe246a7e5cb8bad4961b9860e93b86f67163	$R \geq 0$, $A \geq 0$, Theorem 4.1; CERTIFIED_PROVED
liu9-h2-boundary.json (liu9_h2_boundary_layer.py) file e4d6d96df3ae435800cef5739fb1ae45b98feb539ab418df116ceb03bd481c9 internal 389804b249cd8b16a47f3c6ab64a70ad4a9afe78097b4a8c6739759804c9eaaa module fcb3ed55e8bd9ad6d2f180f5ffa79f192dacc940a5c7506eca2a85aa612905e6	$\Phi \geq 0$, Theorem 5.6; PROVED
liu9-h2-phi-audit.json (liu9_h2_phi_audit.py) file f2359fa3b3b08bab29965253d777dcf32e4b1aee1e0bacbfb0924b86ac163996 internal a515102c1807849b606bf4a681bd0a67173445b1a22610d397bb779440222cab module b1d9ed3d9da9e7c99c026843b683447809fb38d28dba4c49a087729bf66cb50c	independent audit of $\Phi \geq 0$; AUDIT-PASSED
liu9-h2-general-lift.json (liu9_h2_general_lift.py) file eb4874d39904593bddc3f2ae638d1bad6756a19d4ebf6e06a60427b6bb7ac1ce internal 3ff1e797cd5dbcc4273d9f002b9e540b58829d237b2d344f266e0bea46c02bc8 module b265e726d110098a09c00de331c4433454973ba741966b3018b77afcc225b5c3	Theorem 6.1; PROVED
liu9-h2-mixture-theorem.json (liu9_h2_mixture_theorem.py) file 329f7e2d71af8cd78d1a921c72b9ae4d05b71113134eb3341d69932f19ea24b3 internal b108b781224dff601ffc3f562703d3b620e7da6d89cf6d000b6ccbd06a4773b module 3098a1ca30a0f16582df02e1fb7dfd8fa27970c031167125e696e58ac43dfbab	Theorem 7.1; PROVED

Table 1: The certificate chain. For each artifact: SHA-256 of the file, the internal digest, and the module source. All 21 values are pinned in PROGRESS.md and asserted by the replay driver.

re-runs each of the seven modules twice from the repository root under the same interpreter, requires every run to reproduce the on-disk artifact byte for byte, recomputes the three hashes of every artifact from disk in the module’s own canonicalisation, checks the cross-artifact pins, and asserts every hash against the values pinned in the repository ledger `PROGRESS.md` (which contains the same 21 values as Table 1); the exit status is nonzero on any mismatch. The modules use `python-flint` (Arb ball arithmetic [5]; 400 bits for the certificates of Sections 4–7 and the audit, 480 for the reduction, 320 for its independent re-derivation) and `sympy`, single-threaded; the constants are never re-rounded through floating point, and the sole external inputs are the two audited transcriptions `uc/liu9_objective.py` (Liu’s objective) and `uc/liu9_binding.py` (his equations (87)–(90)), both pinned by the lift and mixture artifacts. Each module carries mutations — deliberate errors in the statement or the certificate — and refuses to certify when any of them is active; all 42 mutations across the chain fire.

What is proved and what is not. Theorems 3.1, 4.1, 5.6, 6.1, 7.1 are PROVED with MACHINE-VERIFIED finite components (identities, enclosures, covers); Theorem 8.3 is PROVED by the elementary argument of Section 8; Theorem 9.1 is MACHINE-VERIFIED. Two blank-context adversarial reviews (one of the Φ certificate, one of the lift and mixture artifacts and of the argument of Section 8) found no soundness defect in the final versions; the first version of the lift artifact was refuted by the reviewer at the truncated constants (Remark 2.5) and repaired. The result has not been refereed outside the repository; that is recorded as the adopted next step. Nothing here concerns the conjectured value $\frac{1}{2}$: Lemma 7.3 shows that c' is the best constant this particular pair of protocols at β^* can give, and improving it requires a different protocol family.

Why interval subdivision alone could not have closed Hypothesis 2. R has a nondegenerate interior zero (Lemma 2.4); a function that touches zero cannot be proved nonnegative by bisection at any depth. The stratification of Section 4 handles the zero by exact algebra plus a Taylor bound, and the identity of Section 5 converts the two-variable $B \geq 0$, which vanishes on the entire diagonal (and on both axes), into the one-variable condition $\kappa \geq 2$ with a uniform margin. Both are what makes the dimension-free reduction of Section 3 usable: Hypothesis 2, a nine-parameter statement, and its general-measure form are decided by two inequalities on the unit square.

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